Legend

- RQ — Research Question
- ? — Branching Questions
- User Studies — Means
- X — Links
- Contribution
- Research Axis (*VR, Both, Proxemics*)
- VR / MR / RW — Study Setup (*Virtual Reality, Mixed Reality, Real-World*)

Challenges of Real-World Human–Drone Proxemic Experiments (e.g., Technical Complexity, Legal Aspect, Participant Safety, Limited Design Space). **Virtual Reality (VR) has the potential to counter these issues.**

RQ2: How to best use VR to investigate Human Drone Proxemics?

How can VR alter people's Human-Drone proxemic behaviours?

Literature Review

→ **VR concepts and characteristics** (Skarbez et al., 2018, 2021; Smolentsev et al., 2017)
→ **Impact on Proxemic Factors:** Distance Perception (Interrante et al., 2006; Messing et al., 2005), Motor Abilities (Fink et al., 2007), Threat Perception (Diemer et al., 2015)
→ **Virtual Studies:** Proxemic (Bailenson et al., 2002), Social Psychology (Ryan et al., 2019)

Informs

VR Guidelines for Human Drone Proxemic Studies

Support / Update

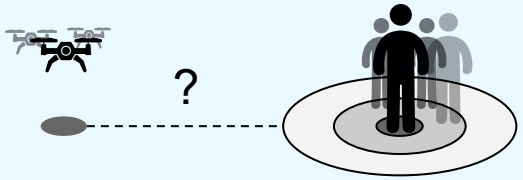
Shape

Motivates

VR Proxemic Studies

VR / Real-World Comparison

Implementation (Proof of Concept)



RQ1: *What motivates individuals' proxemic behaviors around drones and how do they manifest?*

What driving mechanisms ?

What space?

How do people behave around drones ?

User Studies

Explorative

#1 Co-existing with Drones: A Virtual Exploration of Proxemic Behaviours and Users' Insights on Social Drones
Late-Breaking Work Published at CHI 2022 (https://doi.org/10.1145/3491101.3519750)
Full paper Published at the International Journal of Social Robotics (https://doi.org/10.1007/s12369-024-01111-7)

Purpose: 1) Capture people's thoughts and expectations about social drones; 2) Identify considerations for VR proxemic studies; 3) Gather insights on people's reasoning about proxemics with current drones.

VR

Hypothesis-Driven

#2 The Role of Drone's Digital Facial Emotions and Gaze in Shaping Individuals' Social Proxemics and Interpretation
Under Review Transaction in Human–Robot Interaction (2024)

Purpose: 1) Assess the potential social influence of drones using social cues 2) Understand the role and implications of the *communicative proxemic function.*

VR

Hypothesis-Driven

#3 "Do I Run Away?": Proximity, Stress and Discomfort in Human-Drone Interaction in Real and Virtual Environments
IFIP Conference on Human–Computer Interaction - INTERACT 2023 (https://doi.org/10.1007/978-3-031-42283-6_29)

Purpose: 1) Understand the role and implications of the *defensive proxemic function* in HDI. 2) Compare perception and proxemics in virtual vs. real HDI.

VR RW

Hypothesis-Driven

#4 Safety at Stake: How Individuals Task Prioritization Influences Human-Drone Proxemics
Paper Written - Refinement Process with Co-authors Before Submission.

Purpose: 1) Understand the role and implications of the *goal-oriented proxemic function* in HDI and 2) its interaction with the *defensive function.*

VR

Hypothesis-Driven

#5 Drone Noise Level Impact on Cognitive Abilities and Proxemic Preferences.
Analysis Done. Paper Writing in Process.

Purpose: 1) Understand the role and implications of the *arousal regulation proxemic function* in HDI; 2) Assess drone's noise impact and the role of contextual factors.

MR

Literature Review

→ **Human–Robot Proxemics** Meta-analysis (Leichtmann and Nitsch, 2020)
→ **Proxemic Frameworks:** Communicative function (Hall, 1990), Protective function (Dosey & Meisels, 1969), Arousal Regulation function (Patterson, 1976), The specialist model (de Vignemont & Iannetti, 2015)
→ **Cognitive Models:** Situational Awareness (Endsley, 1995); Cognitive Appraisal (Folkman et al., 1986); Biopsychosocial model of challenge and threat (Blascovich & Tomaka, 1996)

Informs

Potential Proxemic Function in HDI

- Goal–Oriented Proxemic Functions
- Defensive Proxemic Functions
- Communicative Proxemic Functions
- Arousal Regulation Proxemic Functions

Motivates

Theoretically Grounded Empirical User Studies on Each Proxemic Functions

Empirical Evidence

Model of Dynamic Human-Drone Proxemics

Supports / Updates

Informs